

19.6 Backus-Gilbert Method

The *Backus-Gilbert method* [1,2], also known as the *optimally localized average (OLA)* method (see, e.g., [3] or [4] for summaries) differs from other regularization methods in the nature of its functionals $\mathcal{A}$ and $\mathcal{B}$. For $\mathcal{B}$, the method seeks to maximize the *stability* of the solution $\hat{u}(x)$ rather than, in the first instance, its smoothness. That is,

$$\mathcal{B} \equiv \text{Var}[\hat{u}(x)] \quad (19.6.1)$$

is used as a measure of how much the solution $\hat{u}(x)$ varies as the data vary within their measurement errors. Note that this variance is not the expected deviation of $\hat{u}(x)$ from the true $u(x)$ — that will be constrained by $\mathcal{A}$ — but rather measures the expected experiment-to-experiment scatter among estimates $\hat{u}(x)$ if the whole experiment were to be repeated many times.

For $\mathcal{A}$ the Backus-Gilbert method looks at the relationship between the solution $\hat{u}(x)$ and the true function $u(x)$, and seeks to make the mapping between these as close to the identity map as possible in the limit of error-free data. The method is linear, so the relationship between $\hat{u}(x)$ and $u(x)$ can be written as

$$\hat{u}(x) = \int \hat{\delta}(x, x') u(x') dx' \quad (19.6.2)$$

for some so-called *resolution function* or *averaging kernel* $\hat{\delta}(x, x')$. The Backus-Gilbert method seeks to minimize the width or *spread* of $\hat{\delta}$ (that is, maximize the resolving power). $\mathcal{A}$ is chosen to be some positive measure of the spread.

While Backus-Gilbert's philosophy is thus rather different from that of Phillips-Twomey and related methods, in practice the differences between the methods are less than one might think. A *stable* solution is almost inevitably bound to be *smooth*: The wild, unstable oscillations that result from an unregularized solution are always exquisitely sensitive to small changes in the data. Likewise, making $\hat{u}(x)$ close to $u(x)$ inevitably will bring error-free data into agreement with the model. Thus $\mathcal{A}$ and $\mathcal{B}$ play roles closely analogous to their corresponding roles in the previous two sections.

The principal advantage of the Backus-Gilbert formulation is that it gives good control over just those properties that it seeks to measure, namely stability and resolving power. Moreover, in the Backus-Gilbert method, the choice of λ (playing its usual role of compromise between $\mathcal{A}$ and $\mathcal{B}$) is conventionally made, or at least can easily be made, *before* any actual data are processed. One's uneasiness at making a post hoc, and therefore potentially subjectively biased, choice of λ is thus removed. Backus-Gilbert is often recommended as the method of choice for designing and predicting the performance of experiments that require data inversion.

Let's see how this all works. Starting with equation (19.4.5),

$$c_i \equiv s_i + n_i = \int r_i(x) u(x) dx + n_i \quad (19.6.3)$$

and building in linearity from the start, we seek a set of *inverse response kernels* $q_i(x)$ such that

$$\hat{u}(x) = \sum_i q_i(x) c_i \quad (19.6.4)$$

is the desired estimator of $u(x)$. It is useful to define the integrals of the response kernels for each data point,

$$R_i \equiv \int r_i(x) dx \quad (19.6.5)$$

Substituting equation (19.6.4) into equation (19.6.3), and comparing with equation (19.6.2), we see that

$$\hat{\delta}(x, x') = \sum_i q_i(x) r_i(x') \quad (19.6.6)$$

We can require this averaging kernel to have unit area at every x , giving

$$1 = \int \hat{\delta}(x, x') dx' = \sum_i q_i(x) \int r_i(x') dx' = \sum_i q_i(x) R_i \equiv \mathbf{q}(x) \cdot \mathbf{R} \quad (19.6.7)$$

where $\mathbf{q}(x)$ and $\mathbf{R}$ are each vectors of length N , the number of measurements.

Standard propagation of errors, and equation (19.6.1), give

$$\mathcal{B} = \text{Var}[\hat{u}(x)] = \sum_i \sum_j q_i(x) S_{ij} q_j(x) = \mathbf{q}(x) \cdot \mathbf{S} \cdot \mathbf{q}(x) \quad (19.6.8)$$

where S_{ij} is the covariance matrix (equation 19.4.6). If one can neglect off-diagonal covariances (as when the errors on the c_i 's are independent), then $S_{ij} = \delta_{ij} \sigma_i^2$ is diagonal.

We now need to define a measure of the width or spread of $\hat{\delta}(x, x')$ at each value of x . While many choices are possible, Backus and Gilbert choose the second moment of its square. This measure becomes the functional $\mathcal{A}$,

$$\begin{aligned} \mathcal{A} \equiv w(x) &= \int (x' - x)^2 [\hat{\delta}(x, x')]^2 dx' \\ &= \sum_i \sum_j q_i(x) W_{ij}(x) q_j(x) \equiv \mathbf{q}(x) \cdot \mathbf{W}(x) \cdot \mathbf{q}(x) \end{aligned} \quad (19.6.9)$$

where we have here used equation (19.6.6) and defined the *spread matrix* $\mathbf{W}(x)$ by

$$W_{ij}(x) \equiv \int (x' - x)^2 r_i(x') r_j(x') dx' \quad (19.6.10)$$

The functions $q_i(x)$ are now determined by the minimization principle

$$\text{minimize: } \mathcal{A} + \lambda \mathcal{B} = \mathbf{q}(x) \cdot [\mathbf{W}(x) + \lambda \mathbf{S}] \cdot \mathbf{q}(x) \quad (19.6.11)$$

subject to the constraint (19.6.7) that $\mathbf{q}(x) \cdot \mathbf{R} = 1$.

The solution of equation (19.6.11) is

$$\mathbf{q}(x) = \frac{[\mathbf{W}(x) + \lambda \mathbf{S}]^{-1} \cdot \mathbf{R}}{\mathbf{R} \cdot [\mathbf{W}(x) + \lambda \mathbf{S}]^{-1} \cdot \mathbf{R}} \quad (19.6.12)$$

(Reference [4] gives an accessible proof.) For any particular data set $\mathbf{c}$ (set of measurements c_i), the solution $\hat{u}(x)$ is thus

$$\hat{u}(x) = \frac{\mathbf{c} \cdot [\mathbf{W}(x) + \lambda \mathbf{S}]^{-1} \cdot \mathbf{R}}{\mathbf{R} \cdot [\mathbf{W}(x) + \lambda \mathbf{S}]^{-1} \cdot \mathbf{R}} \quad (19.6.13)$$